

Software Quality Assurance

Tutorial 12

In this tutorial, you will test with Selenium WebDriver and JMeter. The target application is a sample travel agency web application designed for software testing: <https://blazedemo.com/>

Preparation

1. Open IntelliJ IDEA and click **New Project**.
2. In the New Project window, configure the following settings:
 - **Name:** SQA-Tutorial-12
 - **Language:** Java
 - **Build system:** Maven
 - **JDK:** Java 21
3. Click **Create**
4. Open `pom.xml` and completely replace its contents with the following XML code:

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>

    <groupId>org.example</groupId>
    <artifactId>SQA-Tutorial-12</artifactId>
    <version>1.0-SNAPSHOT</version>

    <properties>
        <maven.compiler.source>21</maven.compiler.source>
        <maven.compiler.target>21</maven.compiler.target>
        <project.build.sourceEncoding>UTF-
8</project.build.sourceEncoding>
    </properties>

    <dependencies>
        <dependency>
            <groupId>org.seleniumhq.selenium</groupId>
            <artifactId>selenium-java</artifactId>
```

```

        <version>4.41.0</version>
        <scope>compile</scope>
    </dependency>
    <dependency>
        <groupId>org.junit.jupiter</groupId>
        <artifactId>junit-jupiter-api</artifactId>
        <version>5.8.2</version>
        <scope>test</scope>
    </dependency>
</dependencies>
</project>

```

5. Reload Maven to Download dependencies

Activity 1

Task

Write an automated test script using Selenium WebDriver (Java) to execute a complete end-to-end booking workflow. Your script must perform the following steps:

1. **Initialize & Navigate:** Initialize the WebDriver and navigate to the target application URL.
2. **Synchronization:** Implement a global wait strategy (Implicit Wait) to ensure the script does not fail prematurely while waiting for elements to load.
3. **Title Verification:** Verify that the homepage has loaded successfully by asserting that the page title is correct.
4. **Find Flights:** Automate the search process by interacting with the dropdown menus to select a departure city and a destination city, then submit the search form.
5. **Choose Flights:** Use Explicit Waits to handle the page transition, then select a flight from the available list by clicking the "Choose This Flight" button.
6. **Purchase Flight:** Wait for the checkout page to load, fill in the required passenger and payment details in the web form, and click the "Purchase Flight" button.
7. **Final Validation:** Verify that the booking was successful by locating the success message on the confirmation page. Remember to properly quit the browser session at the end of the test.
8. **Execution Logging & Observation:** Add `System.out.println()` statements at key milestones to log the test progress in the console. Introduce brief pauses (e.g., `Thread.sleep(2000)`) strictly for educational purposes, allowing students to visually verify the automated interactions before the browser closes.

Preparation

Download and extract ZIP file: https://jmeter.apache.org/download_jmeter.cgi (apache-jmeter-5.6.3.zip)

Activity 2

Task

Build a JMeter test plan from scratch to simulate a user searching for a flight. This activity covers the core anatomy of a JMeter Test Plan.

Travel The World home					
Flights from Boston to London:					
Choose	Flight #	Airline	Departs: Boston	Arrives: London	Price
Choose This Flight	43	Virgin America	1:43 AM	9:45 PM	\$472.56
Choose This Flight	234	United Airlines	7:43 AM	10:45 PM	\$432.98
Choose This Flight	9696	Aer Lingus	5:27 AM	8:22 PM	\$200.98
Choose This Flight	12	Virgin America	11:23 AM	1:45 PM	\$765.32
Choose This Flight	4346	Lufthansa	1:45 AM	8:34 PM	\$233.98

1. **Configure Thread Group (The Users):** Right-click on the Test Plan and add a **Thread Group**. Set the "Number of Threads (users)" to 10 and the "Ramp-up period" to 5 seconds to simulate 10 users accessing the site over a 5-second window
2. **Add Homepage Sampler:** Right-click the Thread Group and add an **HTTP Request Sampler**. Set the Protocol to https and the Server Name to blazedemo.com.
3. **Add Search Flight Sampler:** Add a second **HTTP Request Sampler** to simulate the search action.
 - Set the Method to POST.
 - Set the Path to reserve.php.
4. **Configure Parameters:** In the reserve.php HTTP Request, switch to the **Parameters** tab. Add the data required to complete the flight search:
 - Name: fromPort | Value: Boston.
 - Name: toPort | Value: London.
5. **Configure Timers:** Add a **Constant Timer** to the Thread Group. Set a delay (e.g., 2000 milliseconds) to simulate the "Think Time" of a real human reading the page before clicking the search button.
6. **Add Listeners & Run:** Add a **View Results Tree** or **View Results in Table** listener to gather and display the test execution results. Save your test plan as a .jmx file, click the Run button, and inspect the response data.

FileEditSearchRunOptionsToolsHelp

Test Plan

Thread Group

Home

Constant Timer

Reserve

View Results Tree

HTTP Request

Name:Reserve

Comments:

BasicAdvanced

Web Server

Protocol (http):httpsServer Name or IP:blazdemo.comPort Number:

HTTP Request

POSTPath:reserve.phpContent encoding:

☐ Redirect Automatically

☒ Follow Redirects

☒ Use KeepAlive

☐ Use multipart/form-data

☐ Browser-compatible headers

ParametersBody DataFiles Upload

Send Parameters With the Request:

Name	Value	URL Encode?	Content-Type	Include Equals?
fromPort	Boston	☐	text/plain	☒
toPort	London	☐	text/plain	☒

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